

Data Science for Business (DAT 617)

Spring 2026

Instructor

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Student Meeting Hours (i.e., office hours):

Monday, 4pm – 6pm via [Zoom](#) (passcode: meeting)
Tuesday, 1pm – 4pm in Office

About the Course

Course Overview

This course equips students with a wide array of data science techniques to address diverse business problems. This course explores unsupervised learning and specialized modeling techniques that are central to modern data science applications. Students will move from clustering and dimensionality reduction to natural language processing, time series forecasting, association rule mining, recommender systems, and network analysis and culminating in an introduction to deep learning architectures.

The course is hands-on and application-oriented, emphasizing both conceptual understanding and practical implementation in Python. Each topic will include conceptual discussion of the technique, and an implementation using Python on datasets similar to those used in real-world applications. Students will learn to apply modeling techniques to address an organizational problem. Their learning will be put to the test through guided assignments, a midterm assessment and an exam. In addition, students will have an opportunity to apply analytics frameworks learned to address an organizational problem in a team-based project.

Learning Objectives

1. Understand and Explain key machine learning paradigms beyond supervised learning, including clustering, dimensionality reduction, natural language processing, time series forecasting, network analysis, and deep learning.
2. Apply and Implement advanced analytical techniques in Python to prepare data, build models, and evaluate performance across structured, unstructured, and temporal datasets.
3. Analyze and Interpret model outputs to uncover hidden patterns, relationships, and insights that inform business and organizational decision-making.

4. Integrate and Synthesize multiple machine learning methods—such as text, network, and deep learning approaches—to address complex, real-world problems.
5. Communicate and Present analytical findings effectively through visualizations, reports, and presentations tailored to both technical and non-technical audiences.

Readings

1. McKinney, Wes (2023). Python for data analysis: Data wrangling with pandas, NumPy, and Jupyter (3rd ed.). O'Reilly Media. [Free from Author](#) or [Amazon](#)
2. VanderPlas, Jake (2022). Python Data Science Handbook: Essential Tools for Working with Data (2nd ed.). [Free from Author](#).
3. Géron, Aurelien. (2025). Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems (3rd ed.). [O'Reilly Media](#).
4. James, Gareth, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor (2023). An Introduction to Statistical Learning: With Applications in Python. Springer. [Free from Authors](#).
5. Jurafsky, Daniel, and James H. Martin (2025). Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition. [Free from Authors](#).
6. Bird, Steven, Ewan Klein, and Edward Loper (2021). Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit. [Free from Authors](#).
7. Joseph, Manu and Tackes, Jeffrey (2023). Modern Time Series Forecasting with Python. Birmingham, UK: Packt Publishing. [Amazon](#).
8. Chapman, Chris, and Elea Feit (2019). R for Marketing Research and Analytics. 2nd Edition. Springer International Publishing. [Amazon](#).
9. Banik, Raunak (2018). Hands-On Recommendation Systems with Python: Start Building Powerful and Personalized Recommendation Engines with Python. Packt Publishing. [Amazon](#).
10. Knickerbocker, D. (2023). Network science with Python: Explore the Networks Around Us Using Network Science, Social Network Analysis, and Machine Learning. Packt Publishing. [Amazon](#).

Recommended Python Tutorials

1. Getting Started: [Kaggle](#)
2. [LinkedIn Learning](#) Courses (free for Pace University Students)
3. [DataCamp](#)

Method of Assessment

Assignments [4 x 50 = 200 points]

In the spirit of learning by doing this course will offer you a chance to implement analytical techniques taught in this class by applying the learning to a problem. Assignments contain a series of questions framed as multiple choice, multiple answer or fill-in-the-blank. To answer these questions, you will need to put to practice what was learnt in the associated module by writing code in Python and interpreting results. Each assignment contains two or more sections distinguished with letter suffixes (e.g., Assignment 3a, Assignment 3b). For each section, you will have three attempts. There are four assignments.

In-class Quiz [8 x 12.5 = 100 points]

Lessons learned from class and from completing assignments will be judged through in-class exercises. Exercises must be completed in-class, within a specified period of time and done individually. There are eight graded Quizzes.

Data Science Project [300 points]

A key feature of this class is applying what you have learnt in the curriculum to a real-world dataset. The project requires you to work in a group to perform all of the steps of a typical data analysis project over the course of the semester. Your report is expected to be independently developed by your team. You are free to select a problem, relevant data, and the types of analyses that you will perform. We encourage you to apply your own judgments in deciding how to frame the problems, structure the analysis, and select appropriate methods. This project should be something more than repeating the homework assignments on a different data set; you should aim to make a comprehensive and compelling analysis of an important problem in a domain or application of your choice. The project contains three graded components:

Project Proposal (5%): Identify research problem, find data, and prepare data for analysis.

Final Report (20%): A comprehensive report of analysis, findings, and recommendations.

Presentation (5%): A group presentation highlighting the research problem, analysis conducted and conclusions.

Class Engagement [100 points]

The weekly class session is our opportunity to review the week's material, work through applied exercises, address your questions, and - just as important - get to know each other. It's therefore important to set aside enough time and space (both physical and mental) for our collective task. You will be asked to do all of the following:

- Arrive on time
- Remain in the entire session throughout (excluding any breaks)
- Take part in chats and polls
- Contribute meaningfully to in-class discussions and breakout sessions.
- Respond to instructor questions and requests to share opinions, key takeaways, experiences etc.

The first two classes are particularly important to set the stage for the course, hence missing the first and/or second class will result in loss of class engagement points.

Exam [300 points]

Comprehensive final exam at the end of the semester will be based on all the material covered in class, assignments and readings.

Semester Grades

> = 930	: A	800 – 829	: B -
900 – 929	: A -	700 – 799	: C +
870 – 899	: B +	600 – 699	: C
830 – 869	: B	< 600	: F

General Policies

Schedule

A detailed schedule is available on Classes/Brightspace. This schedule includes week-by-week information on what will be done in class and assignment due dates. This schedule will be adhered to strictly. In case a change is warranted, you will be informed via Classes.

Policy for Missed Assessments

Based on the dates laid out in the schedule at the beginning of the semester, it should be possible for you to plan your schedule so that you don't miss anything. Since assignments may be done over a span of time rather than on a set date at a set time, there will be no opportunity to make up an assignment. There will be 9-10 in-class Quizzes, of which only 8 are graded. While there is no way to make up a missed in-class Quiz, missing just one Quiz will not have an adverse effect on your grade. Since the exam date is also known ahead of time, in general there will be no makeup policy. However, in case of a dire unforeseeable emergency and if I am informed via email, phone, or in person before the exam, an exception may be made. A decision of what qualifies as a dire unforeseeable emergency will be made by me. Finally, requests to take the exam earlier will not be entertained.

Late Submissions

Assessments submitted after the date and time they are due will not be accepted.

Academic Integrity

All students are required to adhere to the statement of academic integrity outlined in the Pace University catalog. Academic integrity infractions can include, but are not limited to, copying and presenting the work of another as your own, collaborating with others on assignments intended to be done individually, copying the work of others during an exam or in-class Quiz, or completing an in-class assessment (including exam, Quiz, or Takeaway) when not in class. You may receive a failing grade in any assessment, exam, or course in which an infraction takes place, and you may be suspended or expelled from the school. When in doubt about what might be considered an academic integrity infraction, the best course of action is to ask me for clarification.

Extra credit opportunities

Opportunities for extra credit, if any, will be made available to all students equally. Individual requests for extra credit will not be entertained.

In-class Policies and Behavioral Guidelines

In-class Policies

Given below are a set of policies that are expected of every student. Consequences of failing to adhere to these policies range from being asked to leave the class to earning an F grade on this course.

- Arrive in class on time and return from breaks in time. Students arriving late tend to distract other students and disrupt class. Students arriving more than five minutes late may not be allowed to attend the session.
- Remain in class until class ends or takes a break. If you leave class during a session, you may not be allowed to attend the next session.
- Do not leave the session during class except during breaks. If you have to leave earlier, it would be better that you not come at all.
- You may not use any device to record either audio or video of the class session. I possess a copyright on the class presentation. Recording any information would be a direct violation of this copyright.
- Use of a smartphone or cell phone in class is not allowed. If you use or even handle a smartphone in class, you will lose all your class engagement points. If this occurs again, you will also drop down an entire grade (i.e., 10%). If this occurs a third time, you will receive an F on the class regardless of past performance.
- You may not do any non-class related work in class. This includes using a desktop, laptop or tablet computer for anything other than taking class notes or doing class-exercises. If you are found doing non class-related work, you will lose class engagement points.
- Demonstrate respect for the professor and student colleagues. Be polite and courteous. Students found being disrespectful or misbehaving will be asked to leave the class session and under extreme circumstances dropped from the course.
- Demonstrate academic honesty and act in accordance with the University policy on academic integrity. Violation of academic integrity standards detailed in the next section will result in failing the class and potentially expulsion from the university.

Behavioral Guidelines

- Work hard. Do not make excuses for failing to complete work in time or point fingers. Do not grade grub.
- Check your Pace email account for class correspondence and check Classes/Brightspace frequently for course-related material.
- Use proper punctuation and grammar in all emails and include your name at the end.
- Come to every class fully prepared with appropriate reading and writing materials and assignments met. Bring the textbook, or any other appropriate materials to class (as required).
- Listen attentively and actively. Take notes on matters discussed in class.
- Participate. Engage in robust, positive and productive class interactions
- Speak clearly, project, be articulate and use proper language
- Dress neatly and wear business attire for key presentations
- Demonstrate strong presentation skills both orally and using effective visual support (i.e. PowerPoint, charts, props)

By continuing in this course, you are agreeing to comply with the Class Policies and follow the Behavioral Guidelines listed above.

Reasonable Accommodations for Students with Disabilities

The University's commitment to equal educational opportunities for students with disabilities includes providing reasonable accommodations for the needs of students with disabilities. To request an accommodation for a qualifying disability, a student must self-identify and register with the Coordinator of Disability Services for his or her campus. No one, including faculty, is authorized to evaluate the need and arrange for an accommodation except the Coordinator of Disability Services. Moreover, no one, including faculty, is authorized to contact the Coordinator of Disability Services on behalf of a student. For further information, please see Information for Students with Disabilities on the University's web site.

Lubin Academic Integrity Statement

Educational institutions should aspire to instill in their students an appreciation for and the practice of ethical conduct. All students are required to adhere to the statement of academic integrity outlined in the Pace University catalog. Academic integrity infractions can include, but are not limited to, copying and presenting the work of another as your own, collaborating with others on assignments intended to be done individually, using unauthorized resources such as an instructor's manual to complete assignments, copying the work of others during an exam, and failing to reference the work of others or creating fake references in your assignments. You may receive a failing grade in any assignment, exam, or course in which an infraction takes place, and you may be suspended or expelled from the school. When in doubt about what might be considered an academic integrity infraction, the best course of action is to ask your instructor for clarification.

Modifications: I reserve the right to make any modifications to the syllabus. You will be notified of any such changes via Classes/Brightspace.

Schedule

Date	Module & Topic	Readings	Submission (to be submitted via Brightspace before the start of class)
Jan 27	Introduction		
Feb 3	Python Basics	McKinney (2023)	
Feb 10	Data Exploration	Ch. 5 – Getting Started with pandas Ch. 6 – Data Loading, Storage, and File Formats Ch. 7 – Data Cleaning and Preparation Ch. 9 – Plotting and Visualization Vanderplas (2022) Ch. 2 – Introduction to NumPy Ch. 3 – Data Manipulation with pandas Ch. 4 – Visualization with Matplotlib	
Feb 17		Lunar New Year – No class	
Feb 24	Clustering	Vanderplas (2022): Chapter 5.11, 5.12, 5.13 Geron (2025): Chapter 8 James et al (2023): Chapter 12	
Mar 3	Clustering	Vanderplas (2022): Chapter 5.11, 5.12, 5.13 Geron (2025): Chapter 8 James et al (2023): Chapter 12	
Mar 10	Dimension Reduction	Vanderplas (2022): Chapter 5.10, 5.14 Geron (2025): Chapter 8 James et al (2023): Chapter 12	Assignment 1
Mar 17	Natural Language Processing	Jurafsky and Martin (2025): Chapter 1, 2, 4, 5, 22 Bird et al (2021): Chapter 1, 2, 3, 5, 6, 7, 11	
Mar 24	Natural Language Processing	Jurafsky and Martin (2025): Chapter 1, 2, 4, 5, 22 Bird et al (2021): Chapter 1, 2, 3, 5, 6, 7, 11	
Mar 31		Spring Break – No class	
Apr 7	Time Series	Joseph and Tackes (2024): Chapters 1 - 9	Assignment 2
Apr 14	Association Rules, Recommendation System	Chapman and Feit (2019): Chapter 12 Banik (2018): Chapters 1, 4, 6	Assignment 3
Apr 21	Network Analysis	Knickerbocker (2023): Chapter 2, 7, 8, 9	
Apr 28	Deep Learning	Geron (2025): Chapter 9-12	Assignment 4
May 5	Project Presentation		
May 12	Exam		